Active InferAnt Stream 001.1 ~ FieldSHIFT-2: Fully Synthetic Dissertations for All-by-All Domains

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all right hello and welcome it is August 5th 2024 this is active infer ant stream dou 1.1 new stream series just dropped and this is where we're headed today we're going to be doing linguistic analysis on fully synthetic all by all dissertations we're going to go into a lot more detail on what that means but just quickly at the top level each of these points in a principal component semantic space is a written dissertation shifting from one field to another following on on the lab of Mike Lev at all who who worked on the field shift one this is field shift 2 it does quite a few different things and for example this data point is the actual term frequency usage Distribution on synthetic negotiation to synthetic Quantum computation I'll talk more about what the synthetic part means this is like a PhD dissertation that's going from negotiation to Quantum computation and here we have similarity clustering across in all by all of different domains so we're going to be going to a lot of different places and ways let's just start with this text file off the bat super excited for this it's been really rapidly developing I'm using cursor 0.39 came out just a few days ago even better than the prior ones so let's start this off right I'm going to push 3,000 files to a major get release this is going to take a few minutes once it pushes push that now all right once that pushes then I'm going to come over to zenodo and make a new release on GitHub so first just wanted while that's loading you see it on the top of the page so noo makes it super easy to connect GitHub repos via zenodo who offers dois digital object identifiers so active inference is right here once the GitHub push is completed then we'll make a release of the repo and then that will get a new dii and archive it on Zoda so that's really awesome so just a few different tools using zodo for publishing for Di GitHub for code and let's reload it see if we're in one minute ago active infant stream1 so now let's make a new release this is going to be number 0 1.1 needs a tag 1.01 during the stream publish it 1.1 is published reload on zenodo and it'll probably take a few minutes but that's pretty cool because then it'll also have the DOI coming from zenodo and the archiving so maybe we'll check back there later okay let us go to these are it's in the background it's translating the dissertations into Sanskrit and Arabic in the background so hence these new files being added but we're going to get all there okay start with GitHub push done reload and confirm update make the GitHub update we did that and we're using cursor 0.39 came out a few days ago okay for backup there's a page in the active block for inst Cota so it summarizes a little bit about field shift so I'm doing this that's right now August 5th 16 UTC the codes in active infrance just uploaded now what field shift 2 is doing right now is generating thousands of these all by all domain a to domain B and I borrow this term and develop in the same tradition from Mike Lan at all's work so here's the paper here machine learning for hypothesis generation in biology and Medicine exploring the latent space of Neuroscience and developmental bioelectricity O'Brien at all 11 so they uh briefly let's see if the figure is a little bit bigger they briefly describe how AI is playing an important and pervasive role in the scientific process they talk about analogies and isomorphisms between for example the neural synopse and the cellular Junction bio electrically and chemically considered they talk about some even broader quote functional invariance between neuros science and developmental biology so that sounds kind of cool promising of course metaphor here's the BART based domain translator this relates to taking material from One content and then putting it into an encoder gets decoded and here they provide the outline of a gp4 based domain translator so Neuroscience Concepts and developmental biology concepts are juxtapose I'll show what their data looked like in a second and they used that they also had annotators they had several person annotators and they tested how how well were the ideas that it came up with they just showed that a few of the kinds of cases that they tested and then did some preliminary go type analyses looking at like were these hypotheses better than a uniform random distribution so that was really inspiring and I had shared previously with Mike Lev the sort of Mike lev. py from elsewhere in the research methods folder um so I thought that that was pretty cool let's look at field shift one and then we're going to go through field shift 2 so what field shift one is first I brought the whole paper's plain text in because then I can use this and say what what would field shift one think about this and then here's what their data sets look like so they had one expression so many memories come to mind so many pattern memories come to the body so they have a kind of expression from the Neuroscience domain and then an expression that's analogous deemed to be analogous from the developmental biology frame so I started playing with that and playing with that and playing with that so here's where we we got to I'll start with what are called domains so this is in field shift 2 now domains I have here ATM transactions biology blockchain Buckminster Fuller chemistry cognitive security cooking entomology fabric free energy principle active inference healthc care Hospitality hopa that's the ants the bees and the Wasps the soft flies Logistics mitochondrial biology negotiation Neuroscience prediction matter expertise Quantum computation spatial web traditional wisdom William Blake uh and we can make another synthetic domain so if anybody is in the chat and they want to request a domain I will um see if we can make it on the Fly I'll show what it looks like but while somebody is posting a fun domain to add it can be as narrow it could be entering the third digit of the pin in the ATM transaction or it could be like that corner but not this corner because a domain it's it could be an ism it could be a kind of a person it can be an entity so it's it significantly generalizes on what I had above with the entities but here's what the structure of these files are so the first line with a hashtag is saying what the domain is just its name then there are several dozens to several hundred examples and questions also depending on the situation and how it was generated and all this other things here like a fact so chemistry may have definitions examples all right Vladimir wrote intelligent soft matter excellent I will add this so new file synthetic intelligent soft m matter. MD okay # intelligence soft matter control okay so contrl a contrl k inline editing using Cloud 3.5 Sonet we're in the domain folder new file that was just created in the syntactic style of at cooking so synthetic these are I just want to pick a few of the synthetics at markdown uh markdown transaction at Hop uh synthetic chemistry let's just see if one is enough please write many relevant professional concise statements and questions related to intelligent soft matter so that's why I describe these knowledge bases as synthetic because they're being entirely synthesized in the file I'm not bringing in any transcripts or any factual databases I'm just saying generate likely expressions giving the the topic so it'll take a while I've I've way way rate limited the um cursor so here we go definitions example definition example fact intelligent soft matter often incorporates nanom materials to enhance functionality and responsiveness okay so looks good crl A contrl K please so now it knows exactly the format too please add many relevant examples and questions but the semantics of what are broadly there are already defined so let's just let it do one more pass then this is perfect because it'll um it'll expand our domains from 22 to 23 okay I'll I'll add one more for um Michael lenon then with the 24 we'll push the 24 through the whole uh outline here's here here's the visualization here's what's going to happen first we're going to generate the input domains that's where we're going to have 24 of them in this case then we're going to generate Pro shifted domains these are just going to be like concatenated text files that will get sent off to the llm it it doesn't have to be its own script or anything like that but what the shift domains. py does is it's going to send this concatenated list of all by all domain shifts so basic basically prompt then domain a then domain B shift domains is going to send that to open API open AI API get back a shifted domain the shifted domain is going to get prompted again to the llm and it will say make a dissertation outline with this structure so then there's a dissertation outline for each all by all then from the outline generate it then this is another llm step then from the draft improve it LOL draft then translate it into a variety of languages and just on the language side this is what that will look like translated dissertations here's the folders Spanish Sanskrit Russian Punjabi Portuguese Navajo it's really interesting what it says for different languages what it will and won't do and then at any of these steps we're basically just talking about a folder of markdown files so we can use a variety of language uh meta analytic methods and use those kinds of landscapes for for our own use okay back to soft matter okay I just wrote please add many relevant examples and questions so these can be you can add in more factual information or specific information so now we're up to we go field shift two just refresh the folder inputs and outputs domain okay let's add the the one that Michael Lennon said just do this and then we'll push both of those on through the all by all non-science ways of knowing in the in the syntactic style of at intelligent soft matter the one we just generated same request here's intelligence of matter in the file structure yeah and this is like gets fascinating with well what does the llm SL pipeline think are likely Salient acceptable safe responses to asking about this or that thank you for the gift up cycle club who wrote stigar G ant Emoji all right 7 2 statements interesting citizen science initiatives we'll just do one more expansion add many more relevant examples and questions all right then we're going to with those 24 domains then we're going to push through the all by all but previously we had 22 domains and so that generated 444 484 pro- shifted domains so let's look at these fil this is just a concatenated file so here's the prompt it's going to say whoa The Prompt is you're an expert follow these steps for the domain transposition there's 12 steps and here's the output okay here's domain go here's the synthetic statements on ATM trans actions all of them then at a roughly equal length here now where domain B is chemistry so then this is going to be ATM isomorphy transposed onto chemistry so it's like the way that people might think about things and the reverse so gripper and gripped push and pull swapping perspective swapping there and back back back and out all right now we have 24 um okay so kind of come along in the cursor and in the file structure so first thing in the terminal just clear LS in the directory with the scripts so we go to the visualization first we had the input domains generate Pro ship domains okay total domains processed 24 to the output directory re okay now we're at 576 Pro shifted domains okay now we're ready to do the domain shift shift underscore domains okay it skips files that Ed had already output so while it's doing that first domain shift awesome while it's doing that first domain shift and it'll report the timing which one it did Dean asked in the chat can you explain field shift in a scale friendly as compared to a scale-free perspective let's revisit that but it's a great question it makes me think of tilt shift photography depth of field so here's what happened the ones that we already had the 484 of the 576 those were skipped so then we got to the First new field shift and we'll we'll look at the script also but it's nothing more than the text file Plus open AI synthetic Neuroscience so Neuroscience Concepts transposed onto intelligent soft matter next new one was so that took 22 seconds they're taking up 19 to 23 seconds now in the inputs outputs reload it so pro- shifted domains is 576 that's our cap but in the shifted domain SD file we're at 47 because we finished three so that is going to keep um plugging meanwhile it was just translating in the background to have more now just we can already start to push we'll run it again later to catch up but we can already take the next step with the 488 shifted domains okay so pull back to visualization what does the shifted domain look like and then how how are we going to use that to write the dissertation outline so here let's look at Logistics applied to hop so every single all by all was just the concatenation of you're a prompt expert you're about to do this domain shift here's domain a here's domain B make the shift so here's what the shift yields so deep analysis of domain a and this could be structured with systems thinking Lon colie George Mobis etc etc etc it would be awesome and there's many more structured that's been one of the most surprising learnings overall of the last like uh five days is yeah there's some structured ways to do cognitive modeling claims analysis rhetorical analysis all these kinds of Concepts however especially with cursor in the game there's so much that can already be brought forth with this rough or fuzzy probabilistic blob and hope strategy so domain a deep analysis domain B separate analysis identification of isomorphy isomorphy sorry identif identify isomorphism so similarities and structure between A and B transposition of fundamental elements novel hypothesis and theory ant colony Supply Chain management develop a new lexicon who's on first outline a research agenda Envision Revolution revolutionizing education identify technological innovations anticipate resistance limitations propose inter disciplinary collaborations construct a compelling narrative so those were the 12 sections from The Prompt so that's why these shifted responses are just responding to the structure that was suggested by prompt so now we're going to go from those shifted domains which are just kind of like fact sheet type reports now we're going to go from the shifted domain to the dissertation outline generator dissertation tab outline generator similar where the dissertation output file already exists it skips it now it's going to be writing an outline so now let's go to what let's look at the dissertation outline script all right so first generate Pro shift domains just really fast logging cursor incredible for helping logging and and bugging and debugging here's the concatenation prompt domain a B please prepare yourself to make a shift domain here's domain a here's domain B for all domains that outputs that concatenation okay now we go to the out the outline generator Dean asks are those AB isomorphisms assumed stable it's a great question put it this way if you didn't change a text file and you ran the same computer it would give probabilistic ically within that envelope the same quote results which may or may not reflect how things actually were of course you can make synthetic domain you can make custom domains you could do Harry Potter you could make different um domains that that also don't have anything to do with anything so maybe you could say this is this is the platonic solid domain this is just very few Expressions very accurate all right outline generator again logging getting the open AI API key this is kept in in like two ways in the M environment file and llm keys. key both of them are in the git ignore so you can put private and I won't open them on the stream but then it enables the scripts to use the API so you can just change that one file like figure out what your API key is put it into one of these kinds of files and then basically there's a way to do it so generate the dissertation outline this one I used another method relatively so instead of whereas in the um um in this concatenator I set the prompt as a variable and then just concatenated three variables so there could be different prompts put in there like it could go a lot of different other ways just to at the at the outset because the prompt here is going to sort of direct how everything it's it's more Upstream whereas in the dissertation outline generator now here's the structure of the dissertation by adding many sections even if it only gives like one to three sentences in each of the sections it's still going to have a good length so it just says provide um you're creating a comprehensive groundbreaking PhD dissertation plan for newly shifted domain this domain represents a fusion Etc it does these could be on new lines it doesn't really matter your task is to create a detailed expansive intellectually rigorous dissertation plan that articulates Etc long line using the provided shifted domain description so that's that kind of like fact sheet report like here's how Logistics could apply to himon optra develop a massive dissertation play then here's the structure expected that's it then it concatenates it with the so that fixed prompt plus the um shifted domain that yields shifted domain dissertation outline so now this is starting to get bumped up from 484 to 493 so let's look at one of these new shifted domains outlines so let's look at see if any have been written for the matter yet let's see which ones have just been written most recently okay synthetic Hospitality to synthetic intelligence soft matter so you it this could be a Hospitality lab saying let's go into soft matter could be the soft matter lab saying let's go into Hospitality but look at this one and the other flip so personalization service quality and sustainability from Hospitality the research aims to redefine the capabilities of soft materials creating responsive user Centric designs that adapt to environmental stimuli so background um on the shifted domain novelty overarching questions so again this is the direction of going how can the principles of guest experience be applied to the design and functionality of intelligent soft materials whereas the text file that says intelligent soft matter to hospitality will say what can we learn from intelligent soft matter that will help us in the hospitality questions and situations then it has all the structures okay new theoretical constructs emerging from the ship so this one these are actually sometimes super funny um and and there could be sections like add more puns and memes and everything but responsive experience Theory user interactions with materials can enhance functionality and satisfaction H like the material it's it's not like its properties are just a priori and fixed they're interactive so there's a a relationship with the responsibity materials quality assurance Model A framework for assessing intelligent materials based based on user experience metrics again let let in the All by all seman uh landscape let let that be the starting point but if there's a if it's a variant or if it's like another way to go then that's just the starting point so that just gets the conversation going um methods ethical personalized soft matter quality metrics for materials smart integration sustainable design principle interdisiplinary implications practical applications future research conclusion let's look at another recent one okay entomology to nonscience ways of knowing dissertation plan how can the principles of Entomology inform personal development Community resilience what new theoretical constructs emerge from the intersection of Entomology and non-science ways of knowing in what ways can this shifted domain influence educational practices societal structures metamorphosis as transformation social structures in community communication models ecological wisdom Etc thank you Michael and Dean great comments All right so back to the visualization so that was generating the outline next dissertation generator. py same logic skipping the files that already existed let's look at what the dissertation looks like so shifted dissertations 484 that was the starting number so these were the ones that were there earlier how about cooking to free energy principle active inference by analyzing how flavor profiles cooking techniques and culinary Fusion can be understood through predictive coding and variational free energy this research chics to redefine culinary arts and enhance the dining experience how can culinary practices be modeled as adaptive systems that minimize free energy what role do predictive coding and variational free energy play in the development of flavor profiles and cooking techniques in what ways can this interdisciplinary approach influence culinary education Etc new theoretical constructs culinary systems as adaptive systems the role of sensory feedback okay here 36 seconds it also includes the self it didn't exclude the the a shift to a so that's actually another informative Loop about about what's there all right but now we have some new ones refresh okay Neuroscience to non-science ways of knowing the significance of This research lies in its intive approach to knowledge integration challenging traditional boundaries between science and non-science by leveraging insights from Neuroscience etc etc etc how can principles from Neuroscience be applied to understand and enhance cultural practices how what are the implications of cultural neuroplasticity for Community resilience and adaptation how do cultural narratives function as symbolic Messengers that influence emotional states and behavior havior what interdisciplinary methodologies can be developed to facilitate this integration of knowledge history ronal non-science ways of knowing new theoretical constructs cultural neuroplasticity Theory maybe it's drawn from somewhere maybe it's just a synthesis symbolic transmission hypothesis whatever is in the schema for the dissertation it will write something with that output there's a few situations where where especially in Translation and things we'll say oh I don't really do that okay let's let's just carry carry on um in the pipeline then that that's the draft so this should maybe be dissertation draft generator but obviously it's all a draft so the dissertation improver okay so it's it you can open a new terminal window if you want to have another open um edge API but um let's um just terminate the one that's Shifting the domains so that's how many we'll we'll just carry forth okay all right dissertation improver skips the ones that are already written it's going to now it's hitting new ones that are also getting pumped in from here this could be coordinated like a lot uh better all right dissertation improver okay this could be iterative there could be different rounds of of commentary so that was again one of the most shocking SL interesting things was um uh on on two two different domains so at the structure at the level of sections of a document like structuring modular publishing and all these kinds of efforts and at the level of structuring like agents and reviewers and critiquer and actor Network model so here's where both of those I was surprised on for the document composition there's very good adherence to this to the prompts like I'm using here like every single PhD outline uses the exact one that's requested so you could add more sections and add the token length and everything um add different sections you could request a different kind of document structured with certain subsections that like do or don't make sense with relationship to your your input folder um so yes it's very strong and interesting to pursue the strongly composable document composition however using clear prompts is giving documents with good structure then on this level of like multi-agent science science agents um writing massive writing all these kinds of things um there's a lot of like flow like what if this went to here and that went to here and I think again that's very interesting and yakob smol and I wrote the generative research teams the grts that discusses a lot and it was exciting okay had to active inference agents and other kinds of of intelligences work together ecosystem of shared intelligence um AOS the the active entity onology for science back in the DSi 2022 days all those kinds of directions like let's have some kind of interoperable strong message passing and again I think that is going to be a strong and a vital element but here with the total cost of millions of tokens still under the tens of cense at least using cursor it it's been easier to just do all by all all by all all by all and just filter folders of markdown files but then just do another combinatory explosion and then filter down and also there's an interesting connection there with how that kind of relates to um abductive inference and the two-stroke Engine with generating and then winwing so so that's actually super interesting too because it turns out I'm not saying well if we connected this one to Blake and that in fabric science it's like just blast filter if needed but if it's not even need it's just text files and it's only a few hundreds to hundreds of thousands should be fine so dissertation improver you're a worldclass academic writer task with improving an existing PhD dissertation enhance it in these 10 ways follow these guidelines like only add things make it better make table added hypotheses enhance the language you could you know whatever it happens to be here's the dissertation to improve insert dissertation in the concatenated file please do it now so here's improving dissertation chemistry to soft matter so some of these it's like and again this is what the language analyses show um like in the semantic spaces in their own different ways whether at the syntactic or the semantic excit is like okay chemistry to chemistry or chemistry to Quantum or chemistry to soft matter those might like cluster closer that's the exact kind of thing that we do measure and analyze and this is just a representative image here because you could imagine all kinds of linguistic um analyses there so let's go to the improved dissertations this slightly um um increases their length I didn't analyze the distribution but like they're all kind of like over 20 mostly so shifted dissertations that was the input improved once Let's do let's just look at one that was already oh well blockchain to Logistics everyone knows about that one it's like okay how about ATM transaction to Bucky Fuller by transposing the efficiency accessibility and security of ATM transactions into Fuller's framework of sustainable Design Systems thinking this research aims to develop a novel model for resource management and Service delivery the proposed daxian kiosk it's like new post Fuller term just dropped will serve as a practical application of this model addressing Global challenges through Community engagement in in technology okay Dean says Michael says it's hilarious and yet strangely educational absolutely and and I think another thread to pull on it's like this is simpler it's so different you know catastrophically different than 2014 to 2019 doing my PhD with limited computer field workor less video chatting but it it's it's easier to just handle the folder slash different so it's it's just it's all you know different is different and then Dean says can you build in information mutation at some specific level would that help the wh ifs over the derivative intersection yeah yeah that's like someone just says like this is just like pulling a crank like like you um you're just like moving strings of pasta forward and then they're just going to be all headed for the trash pile but what you could do is you could be like okay like Okay cooking to Logistics that's kind of interesting maybe there's a grant program or there's like a PhD student applicant volunteer who thinks this is like funny and then synthetic cooking Logistics but it's just a text file so now later we're going to translate this but you could change this and just say we're redacting this part you know replacing that one with that part I'm not going to save it but you can change this or you could write another script that says mutate and it just does any kind of mutation to any subset of these text files okay so then back to visualization now it's translated translation again it's so cheap per translation below below sense per operation so here's translation here's the list of languages we're going to look into the folder and just see what it outputs for some of these translated dissertations so by language folder here's Neuroscience to William Blake in Bengali can't read it chemistry onto itself in Hindi okay languages with uh data sets that are large have strong complete translations we can tell just by looking at the file sizes here so for Hindi they're all 34 is here we have a a range which is interesting so here's a smaller one here's very small I cannot provide a translation of an entire PhD dissertation into Navajo here's a longer one translating a PhD dissertation into Navajo while maintaining High academic quality and Technical accuracy is a complex task that requires deep understanding of both languages and the subject matter below is a translation of the provided dissertation into Navajo following the guidelines you specified neuro Neuroscience onto cognitive security so here it's like I won't but then I did 14 here's a 4.3 okay it's going to be hard it's a complex task however I can provide a sample translation of a portion of the dissertation to demonstrate how this can be approached please note that due to the limitations of this platform which one I will provide a translation of the executive summary and introduction sections only so then that's a glimpse into how it could be translated so look just in the file size you can see heterogeneity and and uh uh both sism when there's little information similarly Sami there's a 228 version let's read it I can't do it and then an even smaller version I can't do it but the ones with a lot of training data they have large consistent file sizes so here it is also logging taking one minute less than a cent per translation skip the ones that it had done 72 67 58 all right uh language analysis last thing then we'll just do some other random uh improvements and look at Super chats and everything Michael Lennon says I feel like I'm witnessing a shifting a pushing of the boundaries of what does science knowledge work in the near future thank you Michael thank you I'm super happy to be able to share this uh and for it to be open source and for there to be so many affordances for people to contribute and fork and clone all right language analysis let's well let's just see what we have running right now okay so this now this one is um finished so you you right now the orchestration like you you kind of have to comb over the folder a few times or you could do let it all shift then let it all write the outline Etc or you could add multiple um multi-thread Computing but these kinds of things are what cursor at all are incredible on it's like selecting a piece of code and then just saying contrl K or control shift L saying make this run on N processors and then there are ways that it will do it well okay so from the terminal again clear okay synthetic domains or it could have been real data or any kind of data all these are just text file outputs you can mutate them arbitrarily at every single step no need for agent orchestration just good old bioinformatic style text file input domains but not saying that that there wouldn't be cases where you'd have higher efficiency Etc with more nuanced approaches and different things input domains they were fully synthetic in this case we had start 22 then we bumped it to 24 with Vladimir and Michael's suggestion concatenate prep optional step if you want you can just batch these two shift domains makes the situation report on shift write the dissertation outline according to how you say it is generate the draft from the outline improve it once expand it add unique hypotheses Etc now translate to n languages in their own folder and then last script for now just the language analysis um oh now it's called metaanalysis shifted dissertation all right so it's running in the terminal processing text so this is a local computation this is not using the AI um or open API Etc um you define the input and the output folder so here we're going to be looking at the analysis for the first dissertation draft but you just change the input folder and you can make it look at any folder um the these are just examples of linguistic analyses again I just control a and just like add more relevant interesting visual outputs to each folder or make a new kind of folder or make me something that will reflect my curiosity about X okay so the script on the prior version slash on disk quote disk is here running but now we're also live editing we're we're confirming that the prior version works there kind of like a little Nano git because then if I accept these green edits and just rerun it right away either it'll continue to run or it'll not okay so here's can now it's added networks interactive and dendrograms will they work let's find out scanning over the parts that are already that are just not being touched but there's times where it will just say oh deletes a huge block of code or there's times where it just does things that that move you in circles but you get to see a lot of code and it's a a well structured code to in at least to this nonprofessional programmers uh cursor enhanced way so it's it's a simpler semantic layer whereas programming when it got lost in the in the obscurity of where the bracket is closing it gets really down there okay so I'm just going to close that prior version clear save the script so if if this works then we'll see new things popping up in this analyses dissertations so if if that that will pop up in these new folders but let then here's here's what these folders have now so this is principal component analysis K means clustering Matrix decomposition methods tne not saying these are the most like performance or anything so many visual improvements but just to say those are clustering methods um oh this one will the the the axes may need to be adjusted depending on how many um like terms are in there okay top terms okay top 30 Terms across all documents okay okay this is just the PCA kind of L that we saw before but there's a shading like a 95% density so this could be used like okay what's that one that one that one and that one or like you know and and then then there's the PCA interpretability methods which look like so here's here's okay so here's pc1 and PC2 these are the these are linear decompositions and and variance quote explanation so here PCA component one so this explains the greatest linear fraction of the variance just to kind of pull back one more step on the pca's every time you take a PC you're explaining more variants that's what what it's called it's not in the causal sense it's the correlational kind of explanation here um but you're always going to explain more variants because you can always get some but then you look at the accumulation curve of the explained variance okay maybe you know and then you could either calculate like oh this is the optimum or we only have space for three but it's good to know that it's only 18% um okay so here's one and two each data point is a written outline so written file names can be improved ET so negotiation is highly weighted on this and Quantum is negative so documents that use negotiation negotiators conflict strategies resolution these are to the right on pc1 documents that mention Quantum or to left PC2 which is the y- AIS is um associated with the use of the word Quantum and for example not the word culinary or mitochondrial let like okay yeah Quantum is up and to the right Etc so that's kind of what these plots are showing but then Landscapes and so oh what would be right here what would be like that one okay term loadings let's see if it works okay specific terms here's again Quantum is is low on pc1 high on PC2 negotiation is slightly high on PC2 very far outlier you might get more resolution like on this analysis by by just excluding these three terms but this is just one way to do the analysis okay it's still chugging along but yeah you can extract what are the terms that are in each principal component so that's the kind of semantic space okay let's see while the visualization method um is running so so like here the translations are just chugging along chemistry to cooking in Portuguese chemistry to cooking Bengali now it's just looping through these languages okay all right so to return to this kind of overall one but we'll watch the visual script Run Okay End to End All by all field shifting domains can be synthetic or Etc dissertation outlining expansion into a draft draft Improvement draft translation meta analytic methods these are shown in the visualization but the visualization was just like CR a contrl k make aski art for this I think combining it with the grant methods that that are in another uh Corner uh of this repo here that's one adjacency upcoming developments okay incorporation of the entity and organization cognitive model so just again this was on the sort of how structured does a cognitive model need to be it was like here's fields. py Chris Fields now these are not even real quotes necessarily but they were drawn from these six I I draw the six interviews in here's friston Bucky Fuller Deborah Gordon Etc and yes there's other people thrown in so it's it's it's nowhere near um anything like a structured cognitive model let alone like the kinds of re real speaker sorted transcripts that that many people in organizations have this was just taking a few podcasts with these people and then just distilling it and this this we can go into it like another time where we can add an entity if somebody wants uh they're again they're these condensed quotes but depending on how you prompt it okay giving a vision plotting here but let's see if it output anything else these are these like py pseudo code is quy now they could be interoperable if you had a schema defined maybe PhD student could work on that um but then they can also loose but bringing the entity and saying okay how could these two entities like this organization and and this person or these two people or these two organizations these three concatenate their entity models like we can even do this and then say how could they refract through this shifted PhD dissertation to Grant what if you can compose it it's at least plausible okay entity and organization cognitive models integration of Grant uh proposals per research meod secure funding for PhD projects and just sort of you know more uh explain like I am dot dot dot so we we could concatenate a prompt that's like um explain this shifted um about the dissertation that's applying ATM transactions to cognitive security explain that like it's this kind of event to this person it it never is gonna at least be interesting and funny what it shows okay thank you Michael for your great comments always very insightful okay so then we we could do how would you explain this to dot dot dot and then that generalizes oh 17 18 19 20 but then so what does the model think about how you should explain something and then it's like well let's call multiple models if they're getting so so so available now that's the whole deeper questions about like externalities Etc Grant methods okay just as a brief uh recap well I'll put the uh in the live chat here's the field shift Koda here's the gram method so this is this was from prior uh so this kind of goes through what the grant concatenation methods do I'm not going to go through it on this stream but they're in the same research methods folder okay yeah and and any other things so maybe one we can try one suggestion that somebody has like one in analysis or a new domain or a new entity or like let's just do one random thing [Music] um you can support the project and earmark donations at donate. active Institute future streams oh this will be there's actually a few a few other interesting topics while people are thinking about um what else to say okay uh P3 there are some pretty extensive p3f methods in the repo it'll be a later ACM uh ant stream but check it out if you look at the repo uh it would be awesome to have if if someone sees like an adjacency or if they see something that they can add to like section n any of these kinds of of like Frameworks that could be awesome and then we go oh make a kofka database that sends messages to the Koda API to write these grants and then post it through mbridge to Discord then developing the active inference uh kernel of the infer ant repo so python pmdp Etc Julia RX infer other languages there's several other languages section I okay here was just one sort of thought on the multiply scale science so this is just using just a sort of standard academic ontology here PhD student is like one dissertation contains multiple papers which was how it was uh concisely described to me then a principal investigator has a lab with multiple dissertations overlapping serly organizational scale is to have multiple Labs with within one organization and then above the organizational scale especially in an online context it it is a spaceship earth situation and that's part of this um Grand science spaceship earth Journey um one tiny step for an epistemic forager one giant leap for antk kindes flower poem okay this is going to be a short Tennyson poem so the the kind of background on This was um Buckminster Fuller was on the editor team for This Journal called tsquare later known as shelter and Frank lyd Wright wrote this article about how all may may plant flowers and then I looked that up as a title and found that it came from this uh poem so and then also it's going to be awesome to see how the the cursor can enrich this too okay the flower by Alfred hennyson once in a golden hour I cast to Earth a seed up there came a flower the people said a weed to and fro they went through my garden bow and muttering discontent cursed me and my flower then it grew so tall it wore a crown of light but thieves from over the wall stole the seed By Night SED it far and wide by every town and Tower till all the people cried Splendid is the flower read my little Fable he that runs May read most can raise the flowers now for all have got the seed and some are pretty enough and some are poor indeed and now again the people call it but a weed and then I I did this one time but I'll I'll see where it goes this time explain this in the just discuss this in the context of Open Source it it it said some really funny things but then that the way that kind of connected to the doing and the like learning and the the foraging and then also the planting planting experiments collaborations really proceeding on grants um like not just planting the seed but there's the foraging and and like attending and being attended to and planting and Andor foraging for for having Andor being able to have the seed however that doesn't mean every foraging trip Etc okay so open source you know from from back in the day the seed is dismissed resistance from proprietary software success the the theft of the seat at night might symbolize how proprietary companies sometimes adopt open source principles or code without fully embracing the open ethos people use it anyone can use it the final Stan reflects the reality of the open source ecosystem not all projects or Forks are of equal quality what was once novel may become common place or even dismissed again wow okay thank you all the the Chatters let's let's look at let's do one of the let's do one more kind of permutation let's do the explain like I am X so we just saw the uh outline General erator let's go to one that has a fixed prompt it'll be simpler take the copy rename explainer control shift L go on to the right side it could be done in line too but but just sometimes it's more deliberative on the right side adopt this script to be and explain like I am blank analogous to the list of Target like languages make it EG Target audiences Michael wrote Angus Deon Nobel for econometrics rec warned recently about hidden power dynamics behind Quantified models helping with the interhuman around the tech EG uncovering hidden power or fussing forward through competing interests is key thank you Michael oh yeah if anyone has uh something for in the uh 10 to 30 minutes total let's see how far this explainer goes and then uh anything else or like comments or or questions that people want there okay so it changed Target languages to Target audiences keep the API key here's translate dissertation here's explain dissertation so replace Target languages with target audience renamed translate to explain updated the prompt updated file naming adjusted log message and print statements it's like what you did what accept it okay let's tune the uh well first let's just see if there any other visual that that script halted Shannon diversity indices so this the font of course can be fixed but this is kind of interesting this this this may just be a a random draw but still within the documents some have a higher and a lower Shannon diversity of of term usage so maybe those are more interesting or have more combinations heat map term frequency heat map okay yeah y AIS you know scale could probably be improved but there are some things there correlation matrices interactive nothing output it just halted it just maybe it's too big of a jump networks halted PCA 3D in and out think outside the boat box what box Quantum negotiations this showed the vector that we kind of looked at with the um loadings variance accumulation no Vector just the data like again this is just PCA th this was taken from the grant processing pipeline so it was you could look at the heat maps of different entities top terms just across all documents word clouds didn't output okay let's if there's any target audience uh let's see if we can do just a couple so we get a few across each let's have Furious 5-year-old oh auto complete curious It's just like that with the syntax and auto complete but I I'll edit this down to a few but also anyone should write in the chat and we can include a few creative Target audiences skeptic well studied and highly motivated skeptic college students undecided on their major Michael lenon writes some additional dashboard indicators epistemic forging adjacency oh yeah additional audience types okay yeah I'll wait I'll wait a few minutes okay then what's let's just look at the prompt that it's going to throw and while people write that for a minute okay so now the functions explain dissertation so now here's the prompt template fixed prompt your worldclass educator task with explaining PhD dissertation 2 a target audience maintain accuracy while adopting the explanation maintain Integrity dignity Mutual dignity and accuracy while adopting the explanation to the audience's type of understanding and broader person this adjust language these are this is where it gets triply interesting with how people really think about the topics the model training and this and how it's processed adjust this is just we we won't go into deep editing this but this is just what it comes up with as the likeliest thing that you would say to generalize across explanations as a concept okay thank you Michael let let's do curious 5-year-old okay we can go rich or we can just go more simple and and general I would just well this one PhD student then college student so we have we have 5-year-old how about high school senior you're undecided on whether or where to go to college pH so we have five 17 PhD student Advanced um professional Advanced technical professional okay Python 3 dissertation explainer okay bug that was the the one shot so then here's here's it'll it maybe we need to fix the input outputs let's go from original oh it's not original dissertation so that's that's like a it's just like there it didn't take the file structure into strong account um but in many many cases it does it just that's where so while it's kind of thinking that's just because it's saying from original because I used the word original but then it didn't know which one of these so let's go to make it the shifted dissertations once clear explaining dissertation to a curious 5-year-old boom cursor's gotten better and better in every version truly and uh these are some of the minor things that you know whether it's whether how it's cursor side or the llm side just the the way that it's uh doing web search for best practices and and able to par some of these just Arcane syntax it's incredible all right they are coming out into explain dissertations we'll let a few accumulate okay the goal this script doesn't really work or this overview now needs to ad adopt slightly but the goal of this was to make like let's see also it's this is one other thing is sometimes it helps to manually resync the index that's the vector embedding that cursor is using um under the hood so it won't be able to unindexed code it will not be able to find so I I I even if you do add new files it it it here now it's synced it so now control K let's just do I think control shift l update this to reflect the structure input outputs functions Etc of everything now hundreds of changed files right so that it's processing on that the the the dissertation uh proceeds let's update this so here's another example example little minor limitation but it'll be handled it just pops in and out of the text formatting this is something it's very challenging to resolve actually in general is is in and out of text in text in text especially if it's all plain text see like this is Fantastical that's there's there are no such however these are partially accurate there aren't the explanations so it's like wildly ad mixed combinations of close technical very subtle uh errors on through just like totally believable but also and then just like okay here's how you run the script it's like no it's not reject it okay explanations all right 5-year-old okay okay first it's starting with all right so let's let's do okay the Neuroscience to we're applying Neuroscience methods to biology for a 5-year-old so this is maybe like someone Mike L's lab from that original field shift paper like Neuroscience Concepts onto developmental biology okay what's the big idea understanding how nature and brains work together what's the big idea imagine if nature like a big park with lots of plants and animals Works a bit like a brain this project called the dissertation Etc why is it important questions we want to answer what we found out before found out about brains we found out about biology those are the from the two domains same strategy new ideas nature can change learning from experience neuro Concepts applied to ecosystem but I remember the paper like what can ecosystems learn and it's it's always coming up and it's so it's like stabilize the metaphor raise the water level on the metaphors overall and the kind of T the water table to to to be like this for every all by all how we're going to learn more what we'll discover why it matters what's next let's keep exploring even little emojis okay now let's go to high school it's still in the it's uh still okay let's let's go to PhD Neuroscience to cooking written more in a high school textbook style background significance and Novelty overarching research questions literature review overview of neural networks table one I don't know how you pronounce some of these uh tables in conversation 58 perceptron okay historical development the culinary Arts current state ofth Art here's the gaps and opportunities pretty long this is this is looking more like well this was oh this was for a PhD students so this is more like here's the kind of brief on each of the sections of dissertation Neuroscience to cooking let's see if this one has similar okay this is like sharing uh your your overall outline for like quals something High School senior okay so this is could be the um same one we looked at with cooking but now for the high school senior yeah what if cooking was not just an art but a science because it's Neuroscience to cooking background significance simpler questions but still broadly like the uh structure of the dissertation here's what the chapters are about but with just one sentence instead of like several paragraphs there however it's structured like a dissertation and then yeah okay let's do a few Mills 8102 wrote I love these wild Mash I bet there's plenty of gold in this approach oh so so we're now we're just pushing like kind of trickling the next few through but let's let's go to the shifted once so these this is these are you know the ones that have been improved once in a certain way so let's do Blake applied to Logistics so this one was super fun I again I just wrote make Expressions about William Blake and add about his mythology by examining the par parallels within Blake's emphasis on creativity spirituality and social critique with the challenges faced in modern Logistics the research used to establish a new paradigm termed the blakean model of logistics that prioritizes Visionary approaches to Supply Chain management okay William Blake's life how can Blake's concept of imagination and contraries and form Innovative practices in logistics and then if someone's like well what about this other important concept it's like first off it maybe it would have selected from it second off add that to the original synthetic domain and then that topic will come up in approaches in what ways can a spiritual approach to work enhance ethical considerations in Supply Chain management what Frameworks can be developed to balance efficiency and sustainability in logistics Blake's accurate birth and death because it was included in the factual claims the blakean model of logistics Duality framework balancing competing priorities in Supply Chain management methodology statistic M usually a kind of mixed method like through structured prompt or any of these way more advanced methods access to the literature Etc you could have it be proposing specific hypotheses it could even be writing the code for for the pre-registration and everything sustainability okay the Blake ones are fun um uh let's quickly look at the prediction matter expertise domain just just because it's it's the it's the outlier domain so this is part of a long running uh Dean tickle ISM with the subject matter expertise like the domains and then prediction matter expertise as this kind of like orthagonal Dimension so for this one I did copy in a few of our specific uh papers like about aliceon Bob way finding so there was a few and then I just said just expand on this concept of of prediction matter expertise come up with modotto come up with memes Switching gears like a pme pro when in doubt synthesize it out predicting the future one cross disciplinary inset at a time so questions examples but it it and then so it just invents all these examples so that one is sort of like that's what's that's the the Paradox with the um pme as anme but let's see what one of those might look like okay so this is Health Care principles applied to pme so pme is what needs to be transformed by establishing a robust framework for user experience quality metrics and Tech driven Solutions Solutions Predictive Analytics patient Centric how can we map Health Care situations to to prediction matters health care what how about something with uh with the spatial web cognitive security to the spatial web cooking to the spatial web examining how principles from The Culinary domain can enhance user experience design in digital environments how can what works for restaurants as as places and times and experiences work for spatial web how about prediction matter expertise to spatial web and then again when it's like oh I would have said something different first off we can have many branches of people's models but also then just we edit the source document this seems to highlight just a predictive statistics and adaptability angle but then the language analysis can be like on these and then you can do it on the explanations and and one of one of the interesting and I think helpful ways to think about a few of these operations that are happening is like the principal component analysis on the term frequency uses so let me pull that one up and then so here turn frequencies are being used across within an across documents so in this situation if you had uh X dissertations in the folder and then you're going to translate into a bunch of different languages like l so if you cluster on syntax like term usage then the languages are going to Cluster together because they're using terms that are like exclusive to one language or another character sets even if you clustered on semantics then the dissertation would cluster together across languages which convey in in the case of an accurate translation like similarish semantics so that's basically the difference between syntax and semantics and that's something that's really opened up with these methods is what what I've been showing with the term frequency type analyses syntax based methods they're still very cool and there's actually a lot of information in them however what cursor especially but the language models overall are enabling is just like arbitrary semantic discussion so thank you Michael you know explain at field shift overall to a clam okay Advanced technical here we go Neuroscience to blockchain for advanced technical foundational work proposed integrative M model neurochain adaptive consensus learning smart contracts mix method case studies surveys interviews simulations analytical approaches so here's that's the explanation script is running oh here's the clam you're nestled in your cozy ocean bed and suddenly the water around you starts to change that's a a bit like what field shift does but for researchers studying the world above the waves field shift is a special tool that helps scientists understand how the environment is changing interesting field shift collects a lot of data about the environment like temperature rainfall and plant growth it's like what so that's kind of the funny again the the the huge the huge range of what comes out in different contexts let's look at a few more random dissertations yeah these are the ones that were made more recently but we halted that one let's just look at one or two more okay William Blake oh we we looked at a Blake one already but we haven't looked at a let's look at one Fe see this whole Fe applied to X this is like I hope broadening the scope it's like it's not just a to B for FP to that the the the the immediately available adjacency space is domains to the second power for starter even before imposing other more High dimensional aspects for interdisciplinary from to two but then there's other structures of synthesis other kinds of patterns that that can be emulated but this is just the from to two um so we have whatever expressions and claims we put in about FP and actm that fit in with uh the the llm schema SL length and and attributes th this is the starting point or a starting point for some of those discussions so here's Fe mitob biology transposing the free energy principle to mitochondrial function how can predictive coding and active inference be applied to mitochondrial function different methods like that would relate to the to the actual working it out FP to an ATM transaction it transcends the immediate context of self-service banking it holds broader implications for adaptive systems in technology and user experience design first principles for the ATM FP to Quantum something that that is happening so in the context Andor in the files or it it may F priston how can the concept of PF be applied to optimize Quantum algorithms in what ways do generative models and Fe correspond to Quantum States and Quantum computation what are the implications of active inference for the measurement processes in Quantum systems okay FP to cooking how does predictive coding and active inference apply to flavor development in culinary practices and what ways can generative models inform the evolution of Cuisines and culinary Fusion what educational Frameworks can be developed to integrate FP Concepts into culinary training so um the last edit I want to make is just to update the readme and then push that so that it is accurate for when and then I can update the um description in the code or in the uh YouTube for where all the folders are if anyone has a last comment though please feel free to ask a comment or question position 10 slow request so even when I'm way over 500 oh also here's here's what's what else is amazing going to reload the page I don't think anything will sh I've spent $112 on this whole whole phase of the project first API call I made was on August 1st to gbt 40 I was like okay that's the best model right wrote two of like the two of the first step two first two shifted domains stopped it wait five minutes reload was like oh no well that was that was $3 oh $150 it's like okay and then I I looked up I asked perplexity I was like what are the costs and everything and what's the best one for for for large shock it's like oh GPT 40 it's like okay so that's $2 today and then yesterday when when I went through all the uh I mean when it was going through all the uh like 484 whatever it was that was $3 then activity number of tokens so now now it's hidden to fast but that's even just with a couple python ter here's gbd4 mini that was 11 million 11.7 million tokens for $3 so token's not exactly a word but like this is on the order depending on on everything etc etc I don't know but it's it's it's not it's not 10 10x and it's not a tenth so it's an incredible amount of text processing today 6.8 million $241 $242 and then cursor so cursor I'm paying $20 a month this is just to describe the method methodological and the the kind of financial not the time side but just like cursor $20 a month for the 500 Fast requests but then the slow ones are not that bad so I don't I don't know how it is not with at at all but but it's it's clearly worth um that and and Beyond then the API calls for open AI oh uh some colleagues shared the open router I think it's a great idea to like improve the llm handling make that more flexible be able to do that locally Etc um but just to use what was working now for several dollarss to to create um these all by alls do the shift make the outline write the draft improve the draft translate to arbitrary contrl K add the dis add the explainer into the visualization all right any last comments then I'll end it and push it after this it's like it just says no maybe if I don't call it still okay curious 5-year-old let's see little Blake Neuroscience blockchain chemistry okay Neuroscience to Blake this special study looks at a famous artist and poet named William Blake and how his IDE are like the way our brains talk to each other how can we think about Blake's ideas like a brain what can we learn from mixing brain science and stories here again no well I guess it isn't to be right now check for any last read me but and and the folder structure May slightly change uh these are all just early working folders

I think fonds underneath research underneath methods underneath one this might be a good source for for for longer reference to uh like grants and and to structure that better but already in the research methods folder this is where all the grant methods are but that's kind of Grant meth methods 1.0 so we can make it kind of Grant methods 2.0 field shift Plus+ plus field shift 3 and that would uh maybe with more flexibility like select the promp select the entity combination maybe select the order and then we could just we could get rolling on on many fronts whereas actually like you know in its own meta way this was manual with heavy augmentation for a lot of all by all considered as text files but but but structurally manually DED so it could be distilled and okay I'll just wait one more minute to read anything people WR while I end the stream push to get help it's it's going to be a active project at The Institute for the active infant project so people can contribute acing synchronously with questions issues documentation on the coda pages in The GitHub uh if someone wants to work on this or integrate it with some other way please just contact me we can figure out the right Institute related or or just other Arrangement and add a bunch of pieces to all of this and just part of the beauty of the the code and the open source system is like it it hopefully can ratchet the function like this doesn't do it all but now this is a starting point and the outputs even if you don't run any code at all in the GitHub you can still see the outputs so I'll push it we'll look at the GitHub final and push prepare methods research field shift to everything's in here inputs outputs shift to dissertations crlf you know just whatever you want to search for Lake it'll search on both sides Blake to Neuroscience here it is translated dissertations Sanskrit here it is I don't know cool all right thank you upcycle thank you Mills 818 thank you 8102 thank you Andrew yeah all right well uh we're live the calls go on so I I hope uh people who are interested share it and the the repo and the video and let's do infra ant streams to develop so many of these fractal functionalities of this repo that has long been in the institute's domain Andrew wrote wondering if a separate 10ish minute setup tutorial on how to reproduce the cursor python GitHub integration would be helpful for others to contribute yeah yeah great suggestion like make the issue on the GitHub and then I'll make it an issue and then we'll get it to it cool all right thank bye for